School of Advanced Computing & Information Technology

Department of Computer Science, Gujarat University

MCA123 Data Analytics (Theory)

Assignment - Unit 5 & 6

Probability and Probability distribution and Hypothesis Testing

1. Differentiate between permutations and combinations with examples.
2. How many ways can 5 books be arranged on a shelf if 2 are identical?
3. Define: Mutually exclusive events, Independent events
4. If $P(A)=0.4$, $P(B|A)=0.3$, and $P(B|A')=0.2$, find $P(B)$.
5. Define random variable and distinguish between discrete and continuous types.
6. Define null hypothesis and alternative hypothesis with examples.
7. Differentiate between Type I and Type II errors with a practical illustration.
8. What is the level of significance? How is it related to Type I error?
9. Explain the difference between one-tailed and two-tailed tests with diagrams.
10. Under what circumstances would you reject the null hypothesis?
11. Compare and contrast the p-value approach and the critical value approach in hypothesis testing.
12. How is hypothesis testing different when σ (population standard deviation) is known versus when it is unknown?
13. When do we use a z-test and when do we use a t-test?
14. What is the Normal distribution? State its properties with a bell curve.
15. Explain the importance of Standard Normal Distribution (Z-distribution) and the process of standardization.
16. Describe the Empirical Rule (68-95-99.7 Rule) for the normal distribution.

Unit 5 and 6 (Book name- "Statistics for Business & Economics" by Anderson)

Title	Content	Book Chapter	Example No.
Probability	**basic relationships of probability**	**4**	**15,16,23,33**
Probability distribution	**Binomial, Poisson, Hyper geometric, Normal and standard normal**	**5 and 6**	**Chapter 5 – 1,2,15,16,25,27,38,39,44,46,48 Chapter 6- 11,14,23**
Hypothesis Testing	**Sigma known and unknown case**	**9**	**3,11,12,13,15,23,24,25**